

Sri Lanka Institute of Information Technology

B.Sc. (Hons) Information Technology



Specialization in Information Technology

IT3020 - Database Systems

Laboratory Worksheet 04

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```

3  DROP TABLE clients_obj_tab CASCADE CONSTRAINTS;
4  DROP TABLE stocks_obj_tab CASCADE CONSTRAINTS;
5
6  DROP TYPE client_obj FORCE;
7  DROP TYPE stock_obj FORCE;
8  DROP TYPE investment_tab FORCE;
9  DROP TYPE investment_obj FORCE;
10 DROP TYPE address_obj FORCE;
11 DROP TYPE exchange_varray FORCE;
12 /
13
15 CREATE TYPE address_obj AS OBJECT (
16     street_no    VARCHAR2(10),
17     street_name  VARCHAR2(50),
18     suburb       VARCHAR2(30),
19     state         VARCHAR2(10),
20     pin          VARCHAR2(10)
21 );
22 /
23
24 CREATE TYPE investment_obj AS OBJECT (
25     company       VARCHAR2(20),
26     purchase_price NUMBER(8,2),
27     purchase_date DATE,
28     qty           NUMBER(10)
29 );
30 /
31
32 CREATE TYPE investment_tab AS TABLE OF investment_obj;
33 /
34
35 CREATE TYPE exchange_varray AS VARRAY(5) OF VARCHAR2(20);
36 /

```

1. Implement the following methods for the object tables STOCKS and CLIENTS (of Laboratory Worksheet 04)

- a. A method to compute the yield of stocks: Yield is the dividend expressed as a percentage of the current price (yield = (dividend/current price) *100). Assume that all dividends are for a period of one year.
- b. A method to compute the current price of stocks in US dollars, taking the current exchange rate of AUD to USD as an input parameter (assuming that the prices in the STOCKS table are in AUD).
- c. A method to count the number of exchanges where a given stock is traded.

```

39  -- PART C: Question 1 - Implement methods for STOCKS table (stock_t type)
40  -- Q1(a) Method to compute yield: (dividend / current price) * 100
41  -- Q1(b) Method to compute current price in USD given exchange rate
42  -- Q1(c) Method to count number of exchanges where stock is traded
43
44  CREATE TYPE stock_obj AS OBJECT (
45      company          VARCHAR2(20),
46      current_price     NUMBER(8,2),
47      exchanges         exchange_varray,
48      last_dividend     NUMBER(6,2),
49      eps               NUMBER(6,2),
50
51      MEMBER FUNCTION  get_yield RETURN NUMBER,
52      MEMBER FUNCTION  price_usd(rate NUMBER) RETURN NUMBER,
53      MEMBER FUNCTION  exchange_count RETURN NUMBER
54  );
55  /

57  CREATE OR REPLACE TYPE BODY stock_obj AS
58
59      MEMBER FUNCTION get_yield RETURN NUMBER IS
60      BEGIN
61          IF SELF.current_price IS NULL OR SELF.current_price = 0 THEN
62              RETURN NULL;
63          END IF;
64          RETURN (SELF.last_dividend / SELF.current_price) * 100;
65      END;
66
67      MEMBER FUNCTION price_usd(rate NUMBER) RETURN NUMBER IS
68      BEGIN
69          RETURN SELF.current_price * rate;
70      END;
71
72      MEMBER FUNCTION exchange_count RETURN NUMBER IS
73      BEGIN
74          RETURN SELF.exchanges.COUNT;
75      END;
76
77  END;
78  /

```

Output:

```

SQL> CREATE TYPE stock_obj AS OBJECT (
      company          VARCHAR2(20),
      current_price     NUMBER(8,2),
      exchanges         exchange_varray,...
Show more...

```

Type STOCK_OBJ compiled

Elapsed: 00:00:00.029

- d. A method to compute the purchase value of all the shares of a client (for each share purchased, $\text{purchase value} = \text{purchase price} * \text{qty}$).
- e. A method to compute the total profit on all the shares of a client (for each share purchase, $\text{profit} = (\text{current price} - \text{purchase price}) * \text{qty}$).

```

80  -- PART D: Question 1 - Create client_obj type declaration (methods declared here)
81  -- Q1(d) and Q1(e) will be implemented in the body
82  CREATE TYPE client_obj AS OBJECT (
83      client_id    NUMBER,
84      fname        VARCHAR2(30),
85      lname        VARCHAR2(30),
86      address      address_obj,
87      investments  investment_tab,
88
89      MEMBER FUNCTION purchase_value RETURN NUMBER,
90      MEMBER FUNCTION total_profit RETURN NUMBER
91  );
92  /
93
94  CREATE TABLE stocks_obj_tab OF stock_obj (
95      company PRIMARY KEY
96  );
97
98  CREATE TABLE clients_obj_tab OF client_obj (
99      client_id PRIMARY KEY
100 )
101 NESTED TABLE investments STORE AS investments_nested;

```

Output:

```

SQL> CREATE TYPE client_obj AS OBJECT (
      client_id    NUMBER,
      fname        VARCHAR2(30),
      lname        VARCHAR2(30),...
Show more...

```

Type CLIENT_OBJ compiled

Elapsed: 00:00:00.068

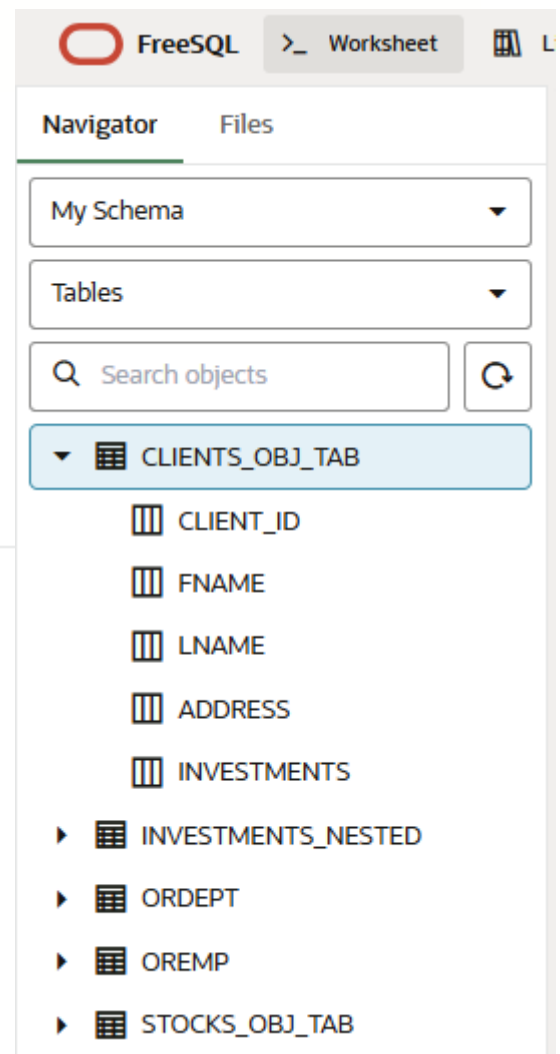
```

SQL> CREATE TABLE stocks_obj_tab OF stock_obj (
      company PRIMARY KEY
)

```

Table STOCKS_OBJ_TAB created.

Elapsed: 00:00:00.029



```

102
103 --Insert stock data
104 INSERT INTO stocks_obj_tab VALUES
105 (stock_obj('BHP',10.50,exchange_varray('Sydney','New York'),1.50,3.20));
106
107 INSERT INTO stocks_obj_tab VALUES
108 (stock_obj('IBM',70.00,exchange_varray('New York','London','Tokyo'),4.25,10.00));
109
110 INSERT INTO stocks_obj_tab VALUES
111 (stock_obj('INTEL',76.50,exchange_varray('New York','London'),5.00,12.40));
112
113 INSERT INTO stocks_obj_tab VALUES
114 (stock_obj('FORD',40.00,exchange_varray('New York'),2.00,8.50));
115
116 INSERT INTO stocks_obj_tab VALUES
117 (stock_obj('GM',60.00,exchange_varray('New York'),2.50,9.20));
118
119 INSERT INTO stocks_obj_tab VALUES
120 (stock_obj('INFOSYS',45.00,exchange_varray('New York'),3.00,7.80));
121
122 COMMIT;
123
124 --Insert client data
125 INSERT INTO clients_obj_tab VALUES (
126 client_obj(1,'John','Smith',
127 address_obj('3','East Av','Bentley','WA','6102'),
128 investment_tab(
129 investment_obj('BHP',12,DATE '2001-10-02',1000),
130 investment_obj('BHP',10.5,DATE '2002-06-08',2000),
131 investment_obj('IBM',58,DATE '2000-02-12',500),
132 investment_obj('IBM',65,DATE '2001-04-10',1200),
133 investment_obj('INFOSYS',64,DATE '2001-08-11',1000)
134 ));
135
136 INSERT INTO clients_obj_tab VALUES (
137 client_obj(2,'Jill','Brody',
138 address_obj('42','Bent St','Perth','WA','6001'),
139 investment_tab(
140 investment_obj('INTEL',35,DATE '2000-01-30',300),
141 investment_obj('INTEL',54,DATE '2001-01-30',400),
142 investment_obj('INTEL',60,DATE '2001-10-02',200),
143 investment_obj('FORD',40,DATE '1999-10-05',300),
144 investment_obj('GM',55.5,DATE '2000-12-12',500)
145 ));
146
147 COMMIT;

```

Output:

```

SQL> INSERT INTO clients_obj_tab VALUES (
  client_obj(1,'John','Smith',
  address_obj('3','East Av','Bentley','WA','6102'),
  investment_tab(...
Show more...

```

```

1 row inserted.

Elapsed: 00:00:00.051

```

```

SQL> INSERT INTO stocks_obj_tab VALUES
  (stock_obj('INFOSYS',45.00,exchange_varray('New York'),3.00,7.80))

```

```

1 row inserted.

Elapsed: 00:00:00.003

```

```

SQL> COMMIT

Commit complete.

Elapsed: 00:00:00.002

```

```

150 CREATE OR REPLACE TYPE BODY client_obj AS
151
152 MEMBER FUNCTION purchase_value RETURN NUMBER IS
153     total NUMBER := 0;
154 BEGIN
155     IF SELF.investments IS NULL THEN
156         RETURN 0;
157     END IF;
158
159     FOR i IN 1 .. SELF.investments.COUNT LOOP
160         total := total +
161             (SELF.investments(i).purchase_price *
162              SELF.investments(i).qty);
163     END LOOP;
164
165     RETURN total;
166 END;
167
168 MEMBER FUNCTION total_profit RETURN NUMBER IS
169     profit NUMBER := 0;
170     curr_price NUMBER;
171 BEGIN
172     IF SELF.investments IS NULL THEN
173         RETURN 0;
174     END IF;
175
176     FOR i IN 1 .. SELF.investments.COUNT LOOP
177         BEGIN
178             SELECT s.current_price INTO curr_price
179             FROM stocks_obj_tab s
180             WHERE s.company = SELF.investments(i).company;
181
182             profit := profit +
183                 ((curr_price -
184                  SELF.investments(i).purchase_price)
185                  * SELF.investments(i).qty);
186         EXCEPTION
187             WHEN NO_DATA_FOUND THEN NULL;
188         END;
189     END LOOP;
190
191     RETURN profit;
192 END;
193
194 END;
195 /

```

Output:

```

SQL> CREATE OR REPLACE TYPE BODY client_obj AS
      MEMBER FUNCTION purchase_value RETURN NUMBER IS
      total NUMBER := 0;...
Show more...

```

Type Body CLIENT_OBJ compiled

Elapsed: 00:00:00.063

2. Answer the following queries on STOCKS and CLIENTS tables. Use methods wherever possible.

- a. Get the yield of all stocks along with their current prices in US dollars, given an exchange rate of 0.74 USD to an AUD. Display the name of the stock, the exchanges where the stock is traded, the stock's yield and the price in USD.

```

196
197 SELECT s.company,
198        ROUND(s.get_yield(),2) AS yield_percentage,
199        ROUND(s.price_usd(0.74),2) AS price_in_usd,
200        s.exchanges
201 FROM stocks_obj_tab s;

```

Output:

	COMPANY	YIELD_PERCENTAGE	PRICE_IN_USD	EXCHANGES
1	BHP	14.29	7.77	["Sydney","New York"]
2	IBM	6.07	51.8	["New York","London","Tokyo"]
3	INTEL	6.54	56.61	["New York","London"]
4	FORD	5	29.6	["New York"]
5	GM	4.17	44.4	["New York"]
6	INFOSYS	6.67	33.3	["New York"]

- b. Find the company name, current price and the number of exchanges for shares that are traded on more than one exchange, using one of the methods.

```

204
203 SELECT s.company,
204        s.current_price,
205        s.exchange_count() AS number_of_exchanges
206 FROM stocks_obj_tab s
207 WHERE s.exchange_count() > 1;

```

Query result Script output DBMS output Explain Plan SQL history				
		Download ▼	Execution time: 0.012 seconds	
	COMPANY	CURRENT_PRICE	NUMBER_OF_EXCHANGES	
1	BHP	10.5	2	
2	IBM	70	3	
3	INTEL	76.5	2	

c) For each client, get the name, and the list of investments with stock name, yield, current price and earnings per share. Use the method for computing yield in this query

```

210 SELECT c.fname || ' ' || c.lname AS client_name,
211        i.company,
212        ROUND(s.get_yield(),2) AS yield,
213        s.current_price,
214        s.eps
215 FROM clients_obj_tab c,
216        TABLE(c.investments) i,
217        stocks_obj_tab s
218 WHERE i.company = s.company;

```

Output:

	CLIENT_NAME	COMPANY	YIELD	CURRENT_PRICE	EPS
1	John Smith	BHP	14.29	10.5	3.2
2	John Smith	BHP	14.29	10.5	3.2
3	John Smith	IBM	6.07	70	10
4	John Smith	IBM	6.07	70	10
5	Jill Brody	INTEL	6.54	76.5	12.4
6	Jill Brody	INTEL	6.54	76.5	12.4
7	Jill Brody	INTEL	6.54	76.5	12.4
8	Jill Brody	FORD	5	40	8.5
9	Jill Brody	GM	4.17	60	9.2
10	John Smith	INFOSYS	6.67	45	7.8

d) Find the total purchase value of investments for all clients. Display client name and total value. Use the method for computing purchase value in this query.

```

220 SELECT c.fname || ' ' || c.lname AS client_name,
221        ROUND(c.purchase_value(),2) AS total_purchase_value
222 FROM clients_obj_tab c;

```

Query result		Script output	DBMS output	Explain Plan	SQL history
  Download		Execution time: 0.03 seconds			
	CLIENT_NAME	TOTAL_PURCHASE_VALUE			
1	John Smith	204000			
2	Jill Brody	83850			

e) For each client, list the book profit (or loss) on the total share investment. Use the method for computing profit.

```
223
224 SELECT c.fname || ' ' || c.lname AS client_name,
225        ROUND(c.total_profit(),2) AS book_profit_loss
226 FROM clients_obj_tab c;
```

Query result			Script output	DBMS output	Explain Plan	SQL history
		Download	Execution time: 0.039 seconds			
	CLIENT_NAME	BOOK_PROFIT_LOSS				
1	John Smith	-8500				
2	Jill Brody	27000				